



YOUR TECHNOLOGY & INNOVATION PARTNER

Advanced AI Training - Session 8

Domain-Specific Applications

AI in Finance



Course Objectives

- Gain an overview of AI's evolution in finance.
- Explore fundamental AI techniques and models.
- Understand the processes for developing AI solutions in finance.
- Identify diverse applications and real-world projects of AI in finance.
- Recognize the challenges and limitations of AI implementation in finance.
- Learn about responsible AI considerations in finance.
- Discover emerging trends and the future direction of AI in finance.

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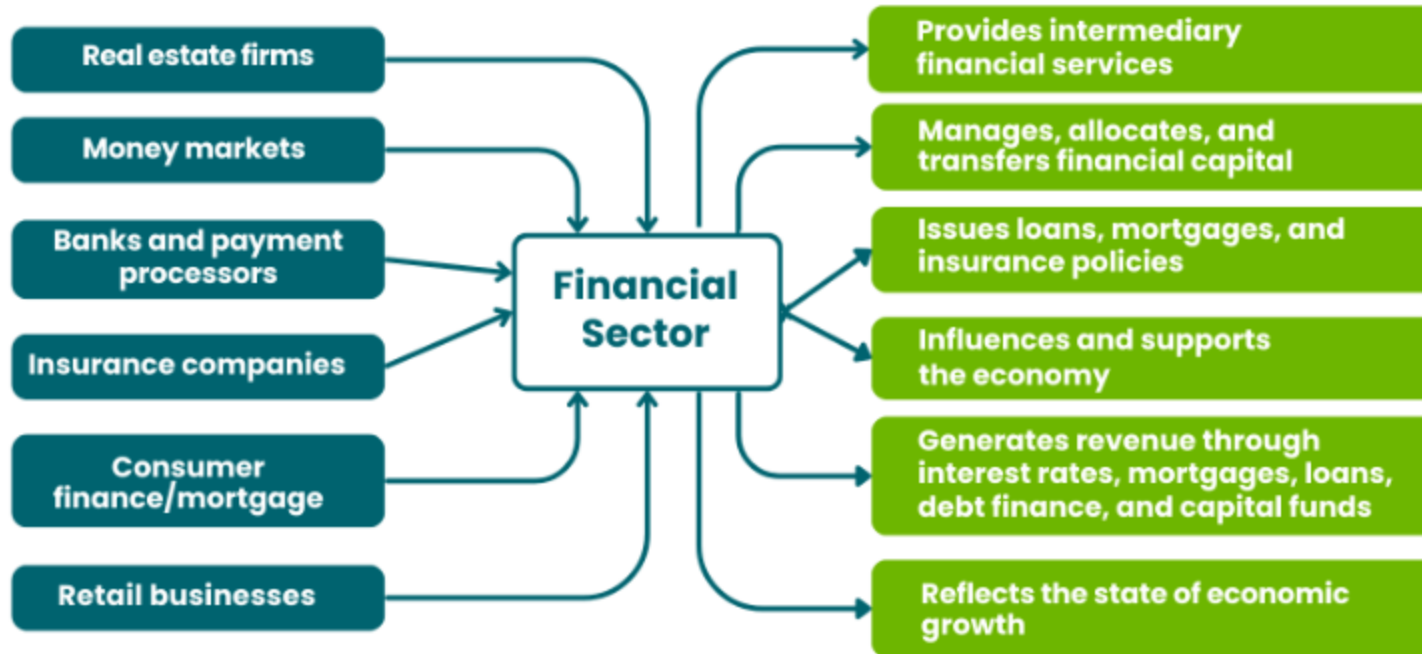


Overview of AI in Finance

Finance

The finance is the part of the economy made up of firms and institutions that provide financial services to commercial and retail customers. This sector comprises a broad range of industries including banks, investment companies, insurance companies, and real estate firms.

What Is Financial Sector?



AI in Finance: Key Milestones

1967

The term “Fintech” first appeared

The term “fintech” itself gained prominence in the article Fin-Tech: New Source of Seed Money

1980s

AI was already being leveraged

AI was applied to areas such as credit scoring, trading algorithms, market analysis, investment, currency exchange, and risk management

1989

FICO credit score

FICO credit score marked a turning point in credit risk assessment

2000s

ML took center stage in AI

Machine learning (ML) becomes integral to finance operations.

2010

Deep learning involved

Deep learning, enabling AI to handle vast datasets and advancing areas like fraud detection, market analysis,....

2025

Widespread Adoption of Generative AI

Generative AI is widely implemented in fintech, especially areas like customer service and financial modeling.

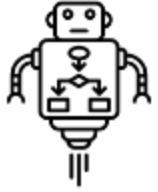
The Data Foundation

	Account and Transaction Data	• Amount, purpose, frequency, time and places of all transaction in history.
	Customer Identity and Profile	• Customer demographics, financial status, loans history, credit history. • Works, income, tax.
	Behavior tracking data	• Device ID, OS version, IP address • Location/GPS data • Browser and session data • App usage logs
	Financial document/invoice	• Invoice, receipt • Financial report, documents
	Market & Financial Data	• Stock prices, crypto prices • Exchange rates • Interest rates • Economic indicators



Finance AI Technologies

AI Techniques



Rule-Based Expert Systems



**Machine Learning
& Deep Learning**



Computer Vision



**Natural Language
Processing (NLP)**



Robotics & Automation



Generative AI

Computer Vision in Finance

OCR

In-house OCR/Open source



[Tesseract \(LSTM\)](#)



[EasyOCR \(CRNN\)](#)



[PaddleOCR](#)

3rd party OCR services



GCP, AWS, Azure

3rd party Vision language model

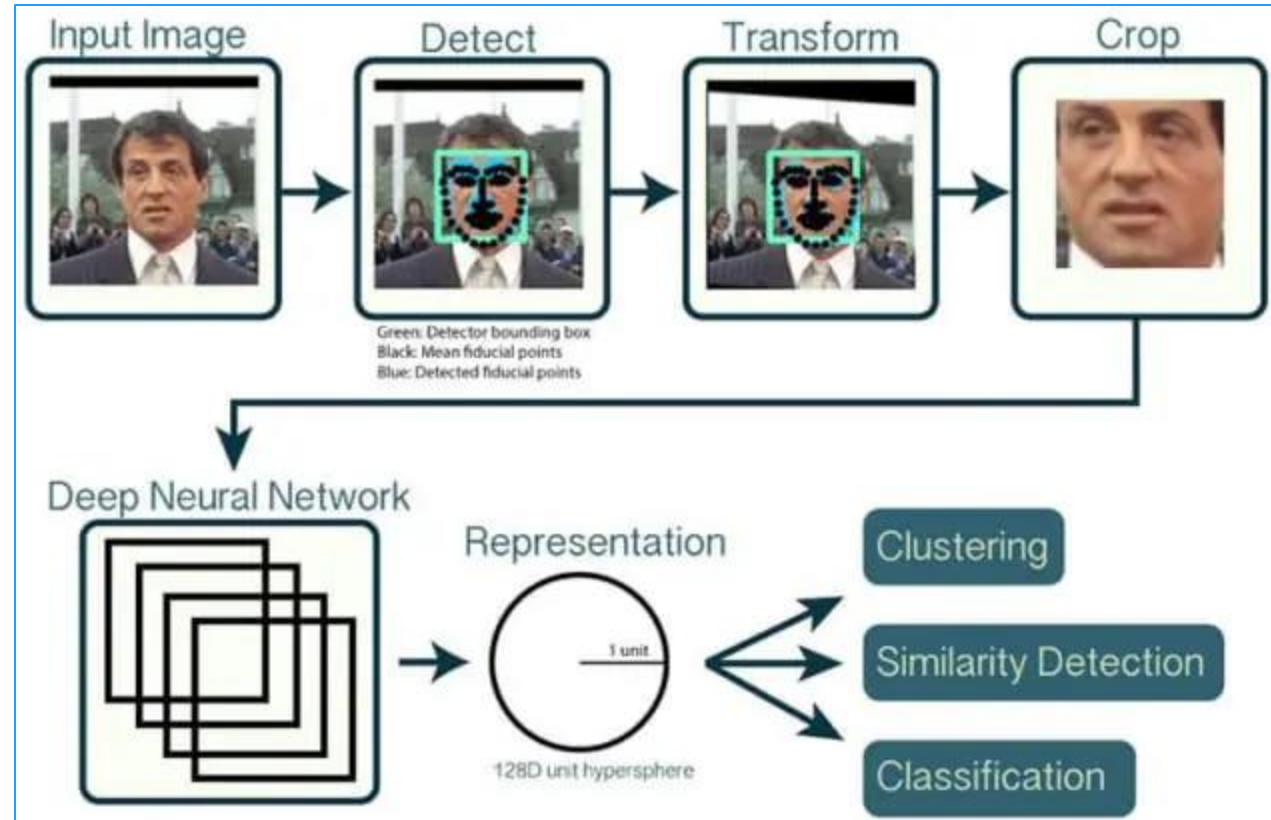


OpenAI GPT 4.1



Gemini 2.5

Face Recognition



Face recognition can be used in many processes such as **KYC**, **Biometric verification**

LLM in Finance

Browse for models



What can LLM do in Finance

Business Event
Detection

Reporting

Risk Analysis

Document
Processing

Entity Recognition

Sentiment Analysis

ESG Analysis

Question Answering

LLM models for financial purposes



FinGPT

is a suite of open-source
medical Large Language
Models (LLMs)

source: <https://huggingface.co/FinGPT>



FinRobot

AI Agent Platform that
transcends the scope of FinGPT,
solution meticulously designed
for financial applications

source: <https://github.com/AI4Finance-Foundation/FinRobot>

InvestLM

Hong Kong's first open-source
model for finance and
investment related GenAI tasks.

source: <https://investlm.hkust.edu.hk/>



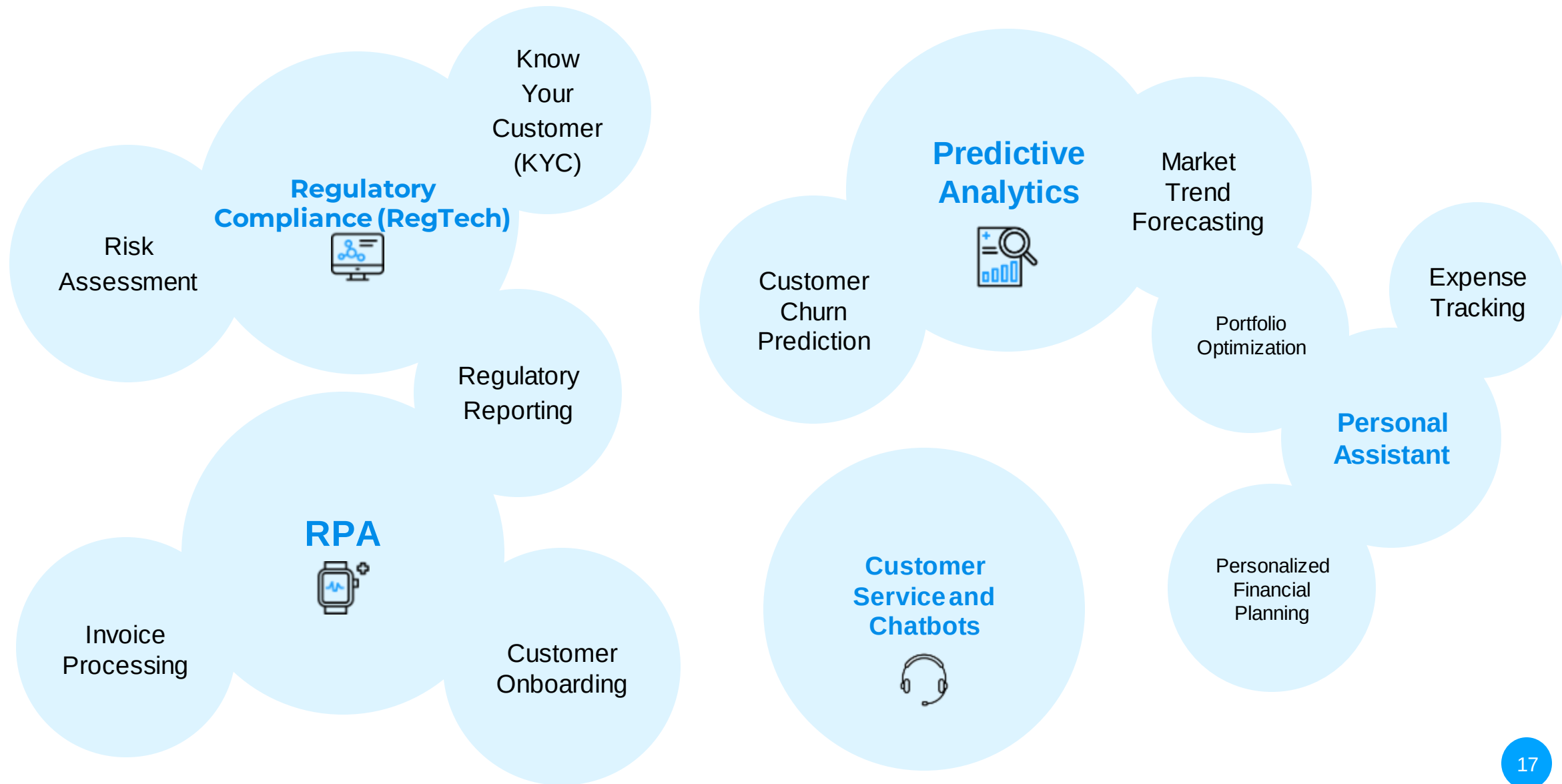
Hong Kong's first open-source
model for finance and
investment related GenAI tasks.

source: <https://github.com/Bavest/fin-llama>



AI in Finance Applications

1. General Use Cases



2. Fraud Detection

Definition

Fraud detection is the process of identifying and preventing fraudulent activities, often involving the analysis of data, transactions, and user behavior to detect anomalies and suspicious patterns. It's a proactive measure used to minimize financial losses, protect assets, and maintain operational integrity.

Techniques

Data Processing

Machine Learning &
Deep Learning

Repositories

Fraud Detection

Credit card

Insurance

Banking & Payment

Data

Identity, transaction time,
amount, category,...

Customer time, age, policies
status, insurance detail, claim
detail,...

Transaction, object and
subject identity,

Used in

- Behavioral Analytics, Chargeback Prevention, Anomaly detection
- Issuance scam, Claims Validation, Risk Assessment
- Money laundering, Scam

2. Fraud Detection

Definition

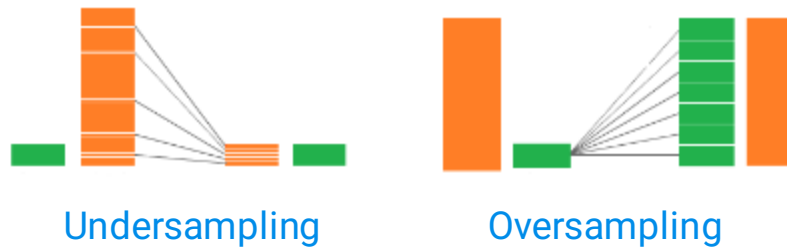
Techniques

Data Processing

Machine Learning &
Deep Learning

Repositories

Preprocessing



Class imbalance:

- Synthetic Minority Oversampling Technique (SMOTE)
- Adaptive Synthetic Sampling (ADASYN)
- Stratified sampling
- Generative Adversarial Networks (GANs)

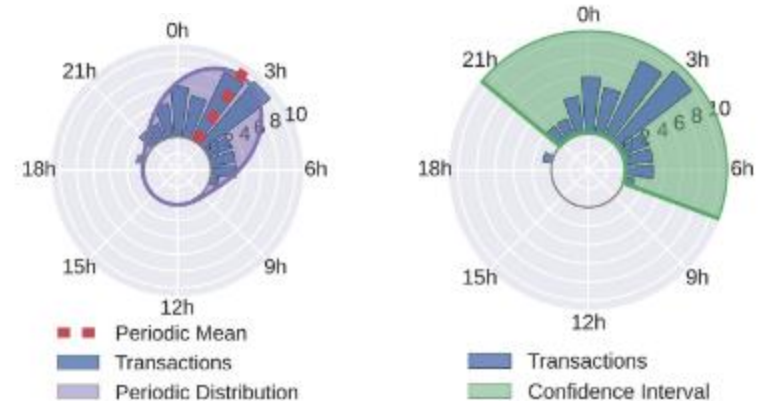
Data Imputation:

- K-NN imputation

Feature engineering

Capture event pattern:

- The time since the last transaction, previous amount of the transaction, previous country of the transaction, number of event in time
- The time of the event in day



2. Fraud Detection

Definition

Techniques

Data Processing

Machine Learning & Deep Learning

Repositories

While aged and less adaptive than deep learning, ML algorithm in fraud detection provide a quick way to archive valuable benchmarks and insights. On the other hand, deep learning achieved a more accurate result and is able to learn from complicated data source without additional data engineering.

Machine Learning

Baseline models:

- Logistic Regression (LR)
- Support Vector Machine (SVM)

Ensemble method:

- Random Forest
- Gradient Boosting (XGBoost, LightGBM)
- Adaptive Boosting (AdaBoost)

Deep Learning

- **Convolutional Neural Networks (CNNs):** Analyze and identity pattern in high-dimensional data such as time-series or transaction heatmap.
- **Recurrent Neural Networks (RNNs):** Helpful for analyzing pattern in transaction history.
- **Transformer:** process all point at once and use self-attention to extract relationship and feature in transactions.
- **Graph Neural Networks (GNNs):** By modelling interactions and relationship between entities as graphs, GNNs are highly effective in fraud detection

2. Fraud Detection

Definition

Techniques

Data Processing

Machine Learning &
Deep Learning

Repositories

Awesome Fraud Detection Research Papers.

A curated list of fraud detection papers from the multiple conferences from 1995 to 2023

IEEE-CIS Fraud Detection

The IEEE-CIS Fraud Detection Competition, many example, data and code/notebook about fraud detection can be found here

Reproducible Machine Learning for Credit Card Fraud detection - Practical handbook

A guidebook and practical handbook for Credit Payment card fraud detection problems

FDB: Fraud Dataset Benchmark

The Fraud Dataset Benchmark (FDB) is a compilation of publicly available datasets relevant to fraud detection ([arXiv Link](#)) and support dataset loading, standardized train-test splits and performance evaluation metrics.

3. Credit Scoring

Definition

Techniques

Repositories

AI credit scoring outperforms traditional models by continuously learning from data, spotting hidden patterns, and adapting instantly to new trends—no human intervention needed. Unlike rule-based systems that age quickly and often miss credit-worthy individuals, AI evaluates more nuanced and complex relationships to make smarter, fairer predictions.

Data

- **Structured Data:**
 - **Credit Bureau Data:** Payment history, credit utilization, outstanding debts, and credit inquiries.
 - **Financial Data:** Bank account transactions, income levels, and employment records.
 - **Demographic Data:** Age, marital status, and occupation.
- **Unstructured Data:**
 - **Alternative Data:** Social media activity, mobile phone usage patterns, and online shopping behavior.
 - **Textual Data:** Loan application notes, customer communications, or social media posts analyzed via natural language processing (NLP).
- **Time-Series Data:** Transaction histories and payment patterns over time.
- **Categorical Data:** Loan types, credit grades, or binary outcomes (default vs. non-default).
- **Numerical Data:** Credit limits, loan amounts, and debt-to-income ratios.

Used in

- **Loan Approval:** Automating decisions for personal loans, mortgages, or credit cards by predicting default risk.
- **Credit scoring:** provide a measurement for real-time risk assessments of a person or organization..
- **Portfolio Management:** Monitoring credit portfolios to detect early signs of default or delinquency.

3. Credit Scoring

Definition

Techniques

Algorithms

LLM implementation

Repositories

Credit scoring

Calculate credit score

Regression problem

LR, KNN, DNN,...

Loan repayment
likelihood

Binary Classification

SVM, RF,
Boosting,...

Determine risk level

Ordinal
classification

RF, Boosting

3. Credit Scoring

Definition

Techniques

Algorithms

LLM implementation

Repositories

The Template for Instruction Tuning Data

query: [prompt + input] **answer:** [output] **choices:** [labels] **gold:** [index] **text:** [input]

Table-based Instruction

Datasets: Australian, Credit Card Fraud, Polish, Taiwan Economic Journal, PortoSeguro

Example instruction tuning data for Card fraud:

prompt: "Assess the creditworthiness of a customer using the following table attributes for financial status. Respond with either 'good' or 'bad'. And all the table attribute names including 8 categorical attributes and 6 numerical attributes and values have been changed to meaningless symbols to protect confidentiality of the data. For instance, 'The client has attributes: A1: 0, A2: 21.67, A3: 11.5, A4: 1, A5: 5, A6: 3, A7: 0, A8: 1, A9: 1, A10: 11, A11: 1, A12: 2, A13: 0, A14: 1.', should be classified as 'good'."

output: "good" **labels:** { 'good', 'bad' } **index:** 0

input: The client has attributes: A1: 1.0, A2: 27.67, A3: 2.0, A4: 2.0, A5: 14.0, A6: 8.0, A7: 1.0, A8: 1.0, A9: 1.0, A10: 4.0, A11: 0.0, A12: 2.0, A13: 140.0, A14: 7545.0.

Description-based Instruction

Datasets: German, Lending Club, ccFraud, Travel Insurance.

Example prompt instruction tuning data for Travel Insurance:

prompt: "Identify the claim status of insurance companies using the records of travel insurance attributes. Respond with either 'yes' or 'no'. For instance: 'A policyholder aged 41 chosen product 'Rental Vehicle Excess Insurance' of the insurance company 'CWT' through sales channel 'Online' to travel to destination 'ITALY'. The type of insurance is 'Travel Agency', with an effective period of 79, and the company recorded the net sales and commission of the insurance as -19.8 and 11.88.', should be classified as 'No'."

output: "no" **labels:** { 'yes', 'no' } **index:** 1

input: A policyholder aged 36 chosen product 'Cancellation Plan' of the insurance company 'EPX' through sales channel 'Online' to travel to destination 'INDONESIA'. The type of insurance is 'Travel Agency', with an effective period of 39, and the company recorded the net sales and commission of the insurance as 28.0 and 0.0."

["Empowering Many, Biasing a Few: Generalist Credit Scoring through Large Language Models".](#)

3. Credit Scoring

Definition

Techniques

Data Processing

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Credit score classification

A popular dataset for credit scoring problems. Example of AI model can be found here

FDB: Fraud Dataset Benchmark

The Fraud Dataset Benchmark (FDB) is a compilation of publicly available datasets relevant to fraud detection ([arXiv Link](#)) and support dataset loading, standardized train-test splits and performance evaluation metrics.

4. Market Analysis

Definition

Techniques

Repositories

Data

Market and Fundamental Data: Includes historical stock prices, trading volumes, financial statements, and economic indicators (e.g., NASDAQ tick data, XBRL-encoded SEC filings).

Text Data: Financial news, regulatory filings, and social media posts processed via natural language processing (NLP) for sentiment analysis.

Use Case

- Sentiment-Based Trading
- Price Trend Prediction
- Risk Management and Volatility Forecasting

What does the candlestick look like



4. Market Analysis

Definition

Techniques

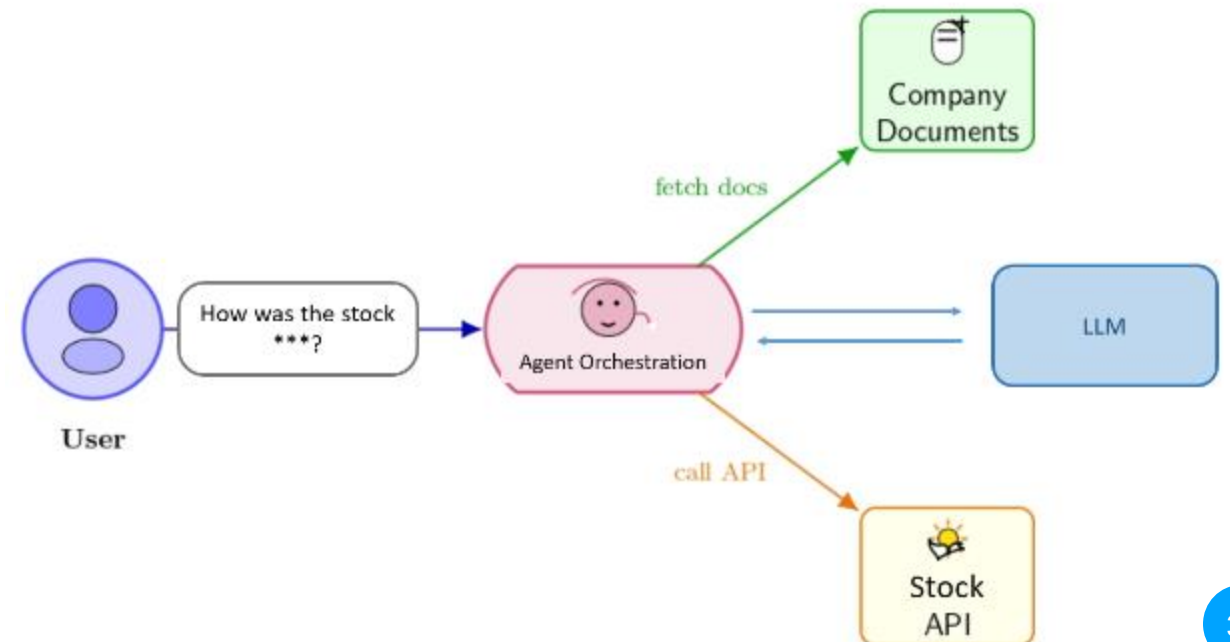
Repositories

Stock/Price Prediction:

- **LSTM**: Effective for capturing long-term dependencies in time-series data, widely used for price prediction.
- **Convolutional Neural Networks (CNN)**: Applied for feature extraction from financial data, such as candlestick charts.
- **Deep Reinforcement Learning (DRL)**: Used for developing trading strategies and portfolio optimization, with models like Deep Deterministic Policy Gradient (DDPG) and Proximal Policy Optimization (PPO).

Market Analysis using LLM:

By connecting to real time news/document and stock API, we can provide live context to LLM, building a market analysis agent that support real time consulting .



4. Market Analysis

Definition

Techniques

Repositories

AI4Finance Foundation

As the world's largest AI finance open-source community Github repositories with many useful projects:

- FinGPT: Open-Source Financial Large Language Models
- FinRL®: Financial Reinforcement Learning. 🔥
- FinRobot: An Open-Source AI Agent Platform for Financial Analysis using LLMs 🚀 🚀 🚀

Huge Stock Market Dataset

The full historical daily price and volume data for all US-based stocks and ETFs trading on the NYSE, NASDAQ, and NYSE MKT. Can find many useful notebook here.

Vnstock

A simple tool to get VN stock and other asset data and news. Can you this to apply AI model with stock and news data.



Finance AI Case Studies

Fintech Center

18+
years

600
engineers

End-to-End
Services

Global
clients

Accounting

Wealth

Capital Markets

Banking

AI in Finance

Payroll

Cashflow
Management

Expenses
Management

Future Plan

Model Portfolio

Post-Trade
Processing

Fund
Management

Installment Loan

Digital
Transformation

Core Banking
Integration

Letter of
Guarantee

Document
Intelligence

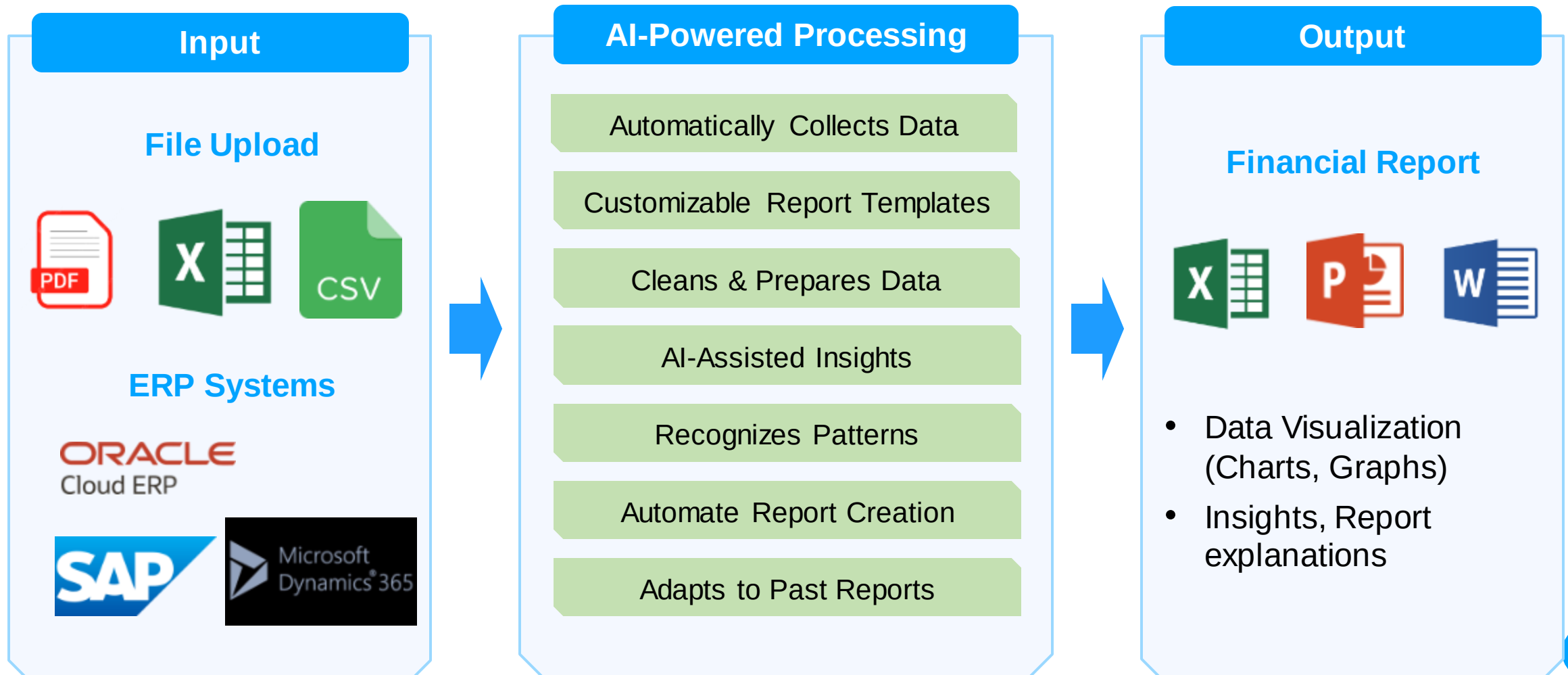
OCR for
E-invoices

AI Generate
Financial Report

AI Finance
Chatbot

Case Study: AI Generate Financial Report

An AI agent to generate financial report on demand



Case Study: Finance Document Intelligence

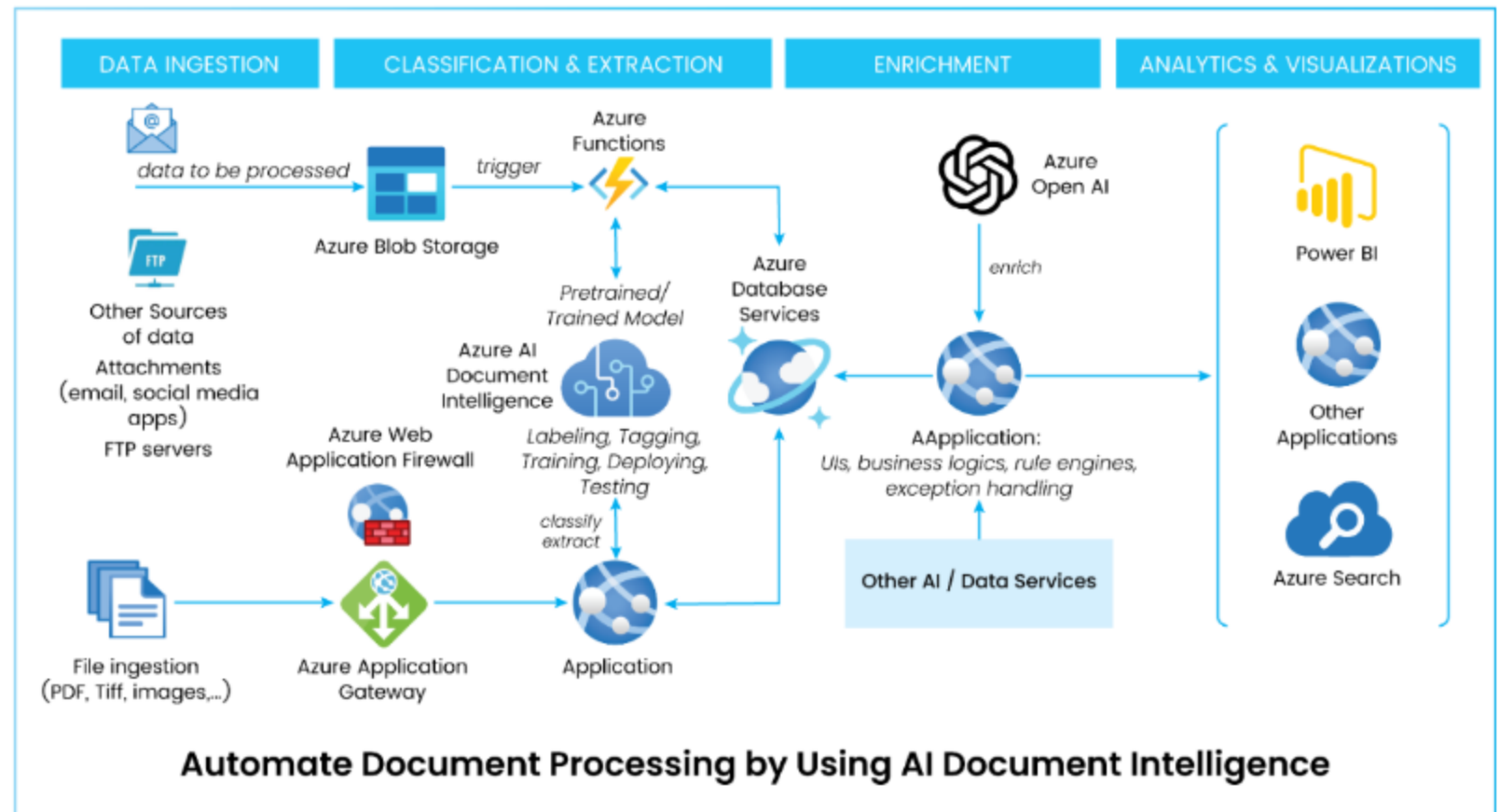
An AI solution based on Azure AI Services, applying advanced machine learning to extract text, key-value pairs, tables, and structures from documents automatically and accurately.

Auto Labelling Data & Adaptability to Change

Custom Data Labelling & Training

Text Detection and Recognition

Data Regions and Compliance



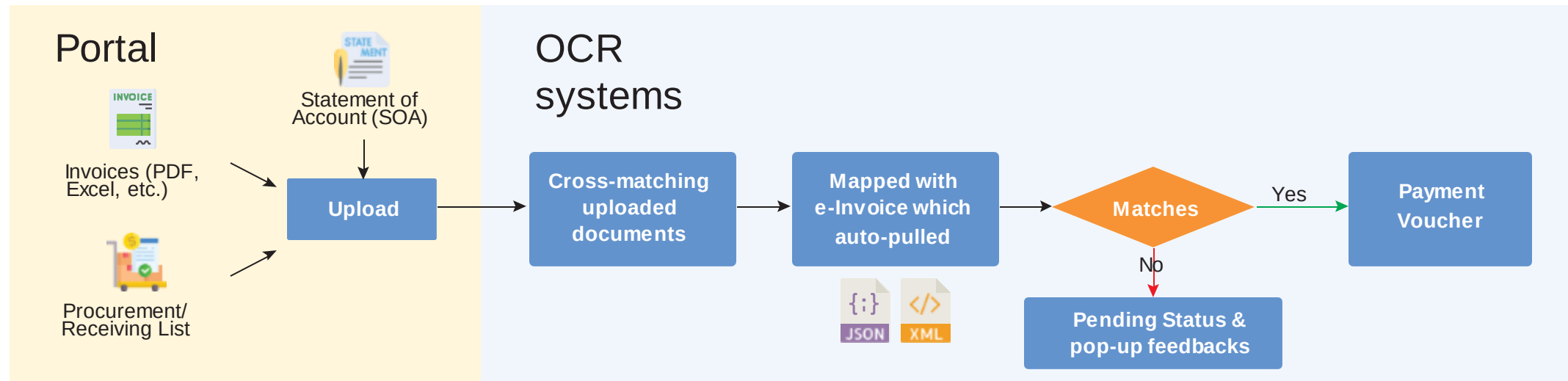
Case Study: OCR for e-Invoice

Data Extraction Portal utilizes Azure AI to manage, classify, and extract text, key-value pairs, tables. Search and highlight specific terms within document files.

AI-powered automation
for document classification & extraction

Search & highlight function
for key financial terms

Supports multiple document types
(Bank Statements, Tax Reports, Invoices)



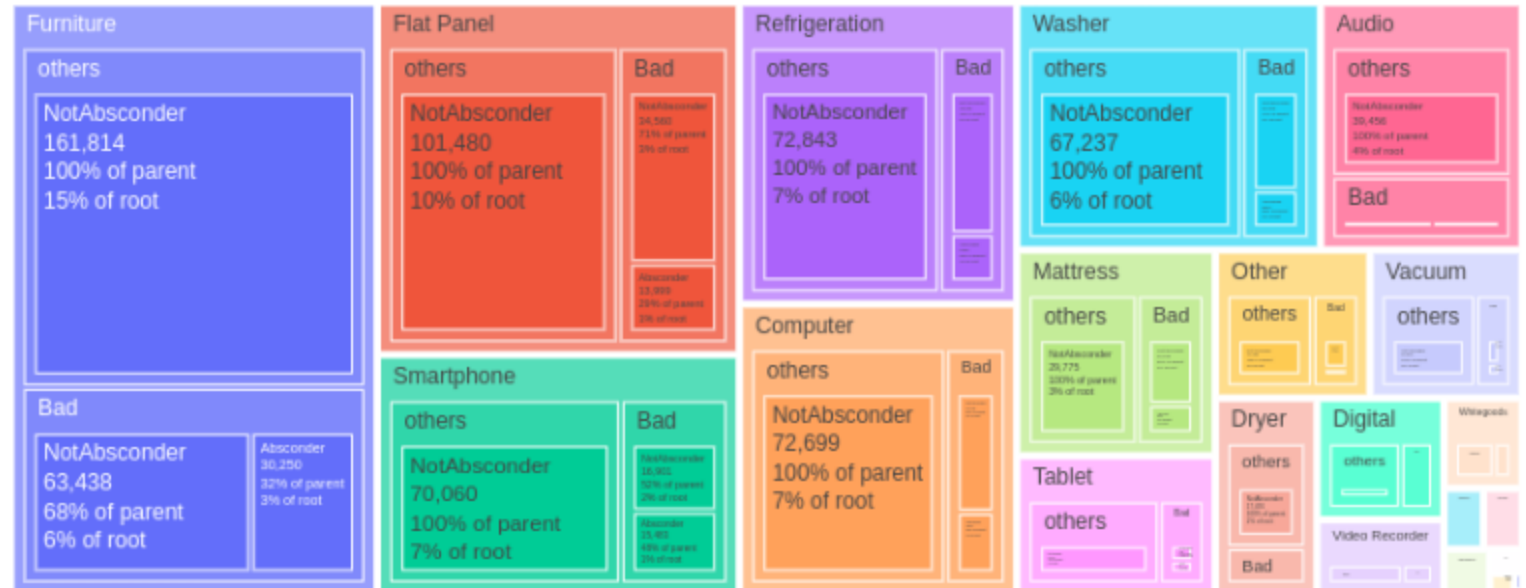
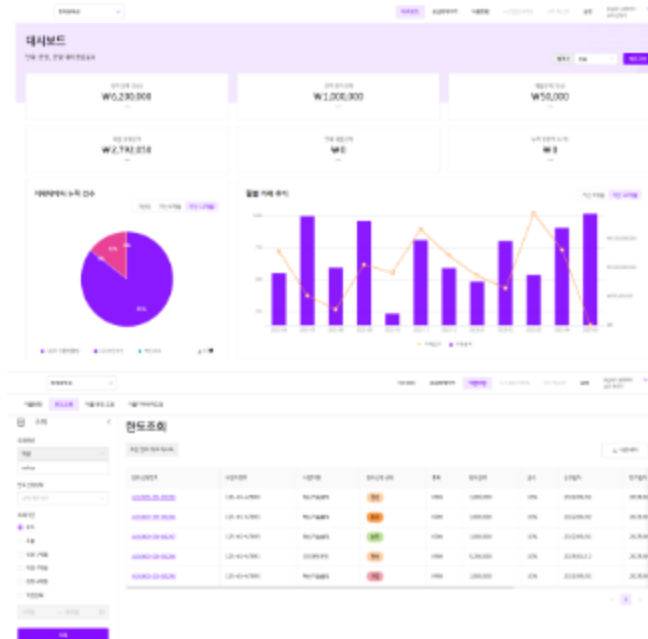
Case Study: Credit Scoring

Create a credit scoring methods based on the given dataset using data analytics techniques and implemented credit scoring rules, suggestions of improvements for the future credit scoring process.

ML-powered Credit Scoring
for a faster and more accurate results

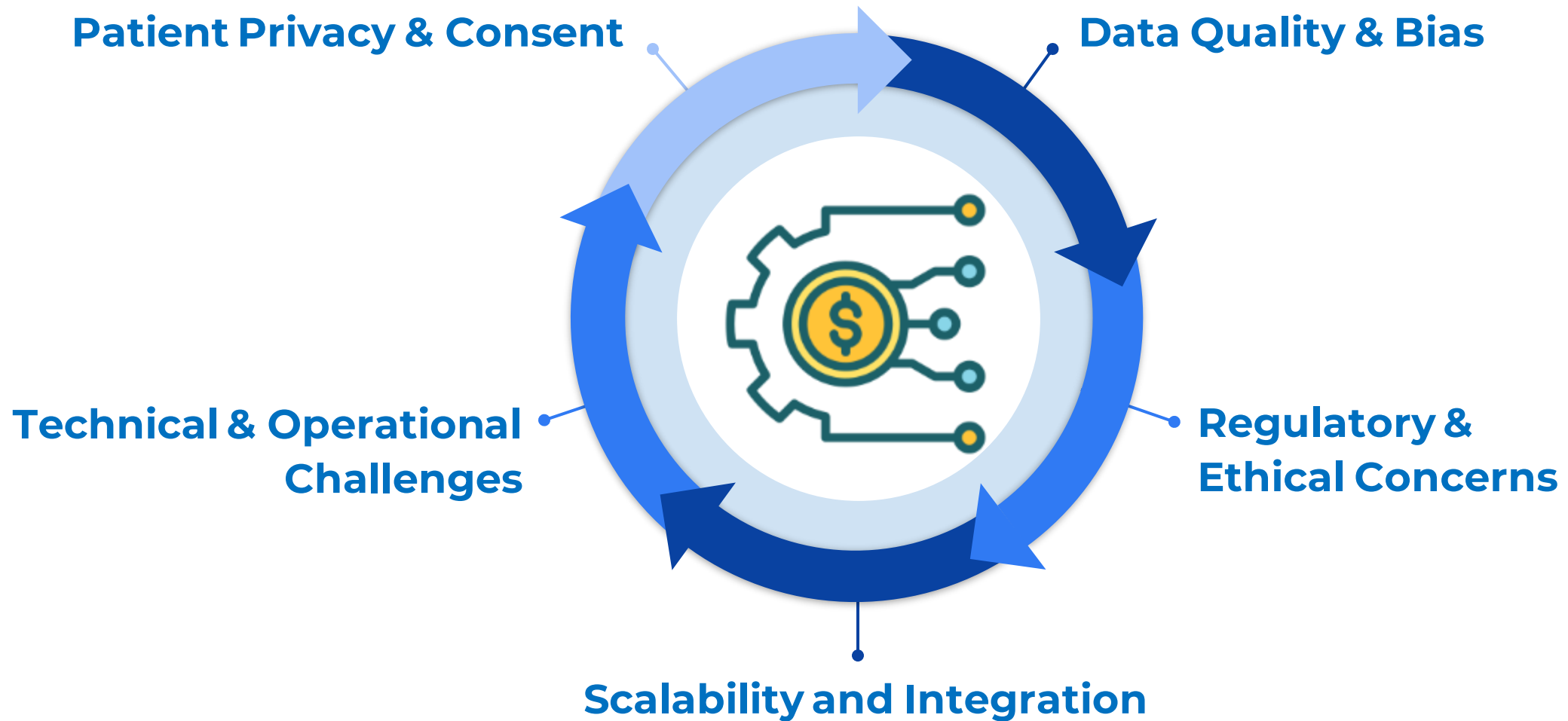
Feature Importance
explain which feature impact the result most

Suggestion for Data & Feature
Based on result, process improvements are suggested to the customer



Challenges and Limitations

Challenges and Limitations



Some Future Scenarios

Inclusive Finance & Alternative Credit Scoring

- AI uses mobile data, utility bills, and behavioral patterns to assess creditworthiness.
- Fintechs are expanding access to underserved populations via mobile-first platforms

WealthTech & Robo-Advisory

- AI-driven portfolio management and tax optimization.
- Tokenized funds and consolidated dashboards democratize investing

AI-Powered Financial Services

- Large Language Models (LLMs) assist with compliance, investment research, and personalized financial planning.
- Explainable AI (XAI) is gaining traction for transparent credit scoring and loan approvals

Decentralized Finance (DeFi)

- Federated learning apply fraud detection to any end-devices.
- Personal AI assistant is customized for every single user

AI RegTech

- FAI and NLP automate audits, detect financial crimes, and streamline compliance reporting.
- Tools like SHAP are being used to explain risk models and ensure fairness

Conclusion

Conclusion

- Artificial Intelligence is revolutionizing the financial sector, driving innovation in areas such as fraud detection, algorithmic trading, credit scoring, and personalized financial services.
- Essential Skills for AI Fintech Solution Development:
 - Using AI ML/DL approaches to solve finance-related problems.
 - Advanced AI Techniques: Expertise in machine learning, deep learning, reinforcement learning, natural language processing (NLP), and generative AI, along with skills in fine-tuning foundation models (e.g., FinGPT) and handling financial data modalities (e.g., time-series, text)
- However, achieving business value with AI in finance requires strategic implementation that addresses challenges like technical complexities (e.g., integrating with legacy financial systems), data quality and bias, and organizational readiness. Not all AI solutions are suitable without careful consideration of these factors.
- Responsible AI Practices: Adhering to ethical AI principles, including fairness, transparency, privacy, security, and regulatory compliance (e.g., GDPR, CCPA), is crucial for building trustworthy and compliant financial systems.



Q&A

